5G Policy Authorization

AM Policy | UE Policy | SM Policy | Call Flows

1. Policy Authorization in 5G — Overview

Policy authorization in 5G is the process by which the network controls and enforces rules governing how data flows and resources are used for a specific application or service session. The **Policy Control Function (PCF)** is the central brain making decisions about QoS, traffic gating, charging rules, and usage monitoring.

Core Network Elements

Element	Role
PCF	Policy decision point — decides all policy rules
SMF	Session management — receives and programs PCC rules
UPF	User plane — enforces rules on actual data packets
AMF	Access & mobility — enforces AM policy, relays UE policy
AF	Application Function — requests QoS for app sessions
UDR	Stores subscriber policy data queried by PCF

Key Interfaces

Interface	Between	Purpose
N5	AF ↔ PCF	Application requests for QoS
N7	PCF ↔ SMF	Policy delivery (PCC rules)
N15	PCF ↔ AMF	AM policy + UE policy
N4	SMF ↔ UPF	PFCP — program packet rules
N36	PCF ↔ NWDAF	Network analytics input

2. Three Types of Policy in 5G

2.1 AM Policy (Access and Mobility Policy)

Controls **how the UE connects and moves** within the network. Governs access restrictions and mobility behavior at the AMF level. Created via the **N15 interface** when UE registers.

Parameter	Description
RFSP Index	Radio Frequency Selection Priority — tells RAN which frequency/RAT to prefer
Service Area Restrictions	Allowed / restricted areas where UE can operate
RAT/Frequency Restrictions	Which Radio Access Technologies are permitted (NR, LTE)
Core Network Type Restrictions	5GC vs EPC preference
Subscribed UE-AMBR	Aggregate Maximum Bit Rate across all sessions

2.2 UE Policy (User Equipment Policy)

Rules sent **directly to the UE** (via AMF as transparent pipe) to control UE-side behavior — which network to select or how to route traffic. Key content: **URSP rules** (which app traffic goes over 5G/WiFi/which slice), ANDSP, WLANSP.

URSP Rule Example:

```
App: YouTube
→ Match:  AppID = "com.youtube"
→ Route:  Use Network Slice SST=1 SD=000001
→ PDU Session Type: IPv4
→ Access Type: 3GPP (not WiFi)
```

2.3 SM Policy (Session Management Policy)

Controls **how a specific PDU session behaves** — QoS, charging, traffic gating, and flow-level rules. This is the most granular policy type, created via the **N7 interface** per PDU session.

Parameter	Description
PCC Rules	Per-flow QoS + charging rules
QoS Data	5QI, GBR, MBR, ARP per flow
Charging Data	Online/offline charging, rating group
Usage Monitoring	Data thresholds and monitoring keys
Gating Control	Allow or block specific flows
Session AMBR	Max bitrate for the whole PDU session

Policy Comparison Summary

Feature	AM Policy	UE Policy	SM Policy
Interface	N15 (PCF-AMF)	N15 → NAS	N7 (PCF-SMF)
Scope	Per UE Registration	Per UE	Per PDU Session
Enforced By	AMF	UE itself	SMF + UPF
Lifecycle	Reg → Dereg	Reg → Dereg	PDU Session lifetime
Key Content	Mobility, area restrict	URSP, WLAN select	PCC rules, QoS, charging
3GPP Spec	TS 23.502/23.503	TS 24.526	TS 23.503

3. Policy Authorization Call Flow

Scenario: Video Streaming App (e.g. Netflix) requesting guaranteed QoS bandwidth.

Phase 1 — PDU Session Setup (Default Policy)

```
UE          AMF           SMF               PCF              UDR
|            |             |                  |                |
|--PDU Sess Establish Req->|                  |                |
|            |--Nsmf_PDU-->|                  |                |
|            |             |--Npcf_SMPolicy-->|                |
|            |             |   Control_Create |--Nudr_DM_Query>|
|            |             |                  |<-Nudr_DM_Resp--|
|            |             |<--SM Policy Resp-|                |
|            |             |  (Default PCC Rules, 5QI=9)       |
|<-----------PDU Session Established (Best Effort QoS)---------|
```

Phase 2 — AF Requests QoS (App triggers policy)

```
AF                              PCF
|                                |
|---Npcf_PolicyAuthorization---->|
|   Create Request               |
|   { AppID: Netflix             |
|     UE IP: 10.0.0.1            |
|     Required BW: 20 Mbps DL    |
|     QoS: GBR / 5QI=4 }        |
|                                |
|              [PCF validates: subscriber eligibility,
|               operator policy, sponsored data, slice]
|                                |
|<--PolicyAuthorization Resp-----|
|   { AuthToken, Authorized QoS }|
```

Phase 3 — PCF Pushes PCC Rules to SMF

```
PCF                             SMF                    UPF
|                                |                       |
|--Npcf_SMPolicyControl_Update--->|                      |
|   Notify                       |                       |
|   { PCC Rule:                  |                       |
|     Rule ID: Netflix-Video-01  |                       |
|     Flow: src=203.x.x.x:443    |                       |
|     5QI=4 (GBR Video)          |                       |
|     GBR DL=20Mbps, MBR=25Mbps  |                      |
|     Gating: OPEN               |                       |
|     Charging: Sponsored }      |                       |
|                                |--N4 Session Mod Req-->|
|                                |  { Install PDR/FAR/   |
|                                |    QER for Netflix }  |
|                                |<--N4 Session Mod Resp-|
```

Phase 4 — RAN QoS Signaling

```
SMF               AMF              RAN               UE
|                  |                |                 |
|--Namf_Comm_N1N2-->|               |                 |
|  { QFI=5,        |                |                 |
|    5QI=4 GBR,    |--N2 QoS Setup--->|               |
|    GBR=20Mbps }  |                |--RRC Reconfig--->|
|                  |                |  (New DRB QFI=5) |
|                  |                |<--RRC Reconfig---|
|                  |<--N2 Setup Resp--|               |
```

Phase 5 — Enforced Data Flow

```
UE  <===== GBR Bearer (QFI=5) =====> Netflix Server
     Video stream with guaranteed 20 Mbps QoS end-to-end
                    UPF monitors usage → reports to SMF/CHF
```

Phase 6 — Session Release

```
AF                    PCF                  SMF               UPF
|                     |                    |                 |
|--PolicyAuth Delete-->|                    |                 |
|                     |--SMPolicy Update---->|                 |
|                     |  Notify (Remove Rule)|--N4 Mod Req----->|
|                     |                    |  (Remove PDR/QER)|
|                     |                    |                 |
|          [QoS reverts to default Best Effort 5QI=9]          |
```

4. AM Policy Creation Call Flow (PCF ↔ AMF)

Trigger: UE Registration

```
UE              AMF                     PCF                 UDR
|               |                       |                   |
|--Registration->|                      |                   |
|                | [Auth + UDM Registration]                 |
|                |                       |                   |
|                |--Npcf_AMPolicyControl->|                  |
|                |  Create Request       |                   |
|                |  { SUPI, GPSI,        |                   |
|                |    AccessType: 3GPP   |                   |
|                |    RAT Type: NR       |                   |
|                |    Serving PLMN, TAI } |                  |
|                |                       |--Nudr_DM_Query--->|
|                |                       |  AM Policy Data   |
|                |                       |<--Nudr_DM_Resp----|
|                |                       |  { RFSP Index,    |
|                |                       |    Svc Area Restrict,
|                |                       |    UE-AMBR }      |
|                |                       |                   |
|                |<--Npcf_AMPolicyControl-|                  |
|                |   Create Response     |                   |
|                |   { RFSP Index: 2     |                   |
|                |     Allowed TAIs: [1,2]|                  |
|                |     Forbidden: [TAI-99]|                  |
|                |     UE-AMBR UL=100M   |                   |
|                |              DL=500M } |                  |
|                |                       |                   |
|        [AMF stores + applies AM Policy for this UE]        |
|<--Registration Accept-----------------|                   |
```

AM Policy Update — Location Change Trigger

```
UE                RAN               AMF                PCF
|                  |                 |                  |
|--moves to new TAI>|                |                  |
|                  |--N2 Location---->|                  |
|                  |   Update        |                  |
|                  |          [AMF detects TAI change   |
|                  |           = Policy trigger]        |
|                  |                 |                  |
|                  |                 |--Npcf_AMPolicy-->|
|                  |                 |  Control_Update  |
|                  |                 |  { New TAI,      |
|                  |                 |    RAT Type }    |
|                  |                 |                  |
|                  |                 |<--Update Resp----|
|                  |                 |  { Updated RFSP, |
|                  |                 |    New restrict }|
|                  |          [AMF applies new policy,  |
|                  |           signals RAN if RFSP chgd] |
```

5. UE Policy Creation (PCF → AMF → UE)

Trigger: UE Registration or PCF-initiated push

```
PCF                 AMF                         UE
|                   |                           |
|--Npcf_AMPolicyCtrl->|                         |
|  UpdateNotify       |                         |
|  { UE Policy:       |                         |
|    URSP Rule 1:     |                         |
|     TrafficDesc:    |                         |
|      AppID=YouTube  |                         |
|     RouteSelDesc:   |                         |
|      SST=1,SD=000001 |                        |
```

```
|      PDUType=IPv4    |                            |
|    URSP Rule 2:      |                            |
|     AppID=Zoom       |                            |
|      SST=2,SD=000002 |                            |
|    ANDSP:            |                            |
|     WLAN for bgrd }  |                            |
|                     |                            |
|<--AMF Ack-----------|                            |
|                     |---NAS: UE Config Update---->|
|                     |   (URSP+ANDSP encoded)      |
|                     |<--NAS: UE Config Complete---|
|                     |                            |
|        [UE uses URSP to decide routing per app]   |
```

How UE Uses URSP:

```
User opens YouTube
↓
[UE checks URSP Rules]
  → Match: AppID = YouTube
  → Route: SST=1, SD=000001 (eMBB Slice)
↓
[UE sends PDU Session Establishment]
  → Requested NSSAI: SST=1 SD=000001
  → AMF routes to correct SMF for that slice
↓
[Video stream flows over correct slice]
```

6. SM Policy Creation Call Flow (PCF ↔ SMF)

Trigger: PDU Session Establishment

```
 UE      AMF       SMF                PCF              UDR        UPF
 |        |         |                  |                |          |
 |--PDU Sess Establish Req-------------->|                |          |
 |        |--Nsmf--->|                  |                |          |
 |        |         |--Npcf_SMPolicy--->|                |          |
 |        |         |  Control_Create   |                |          |
 |        |         |  { SUPI,          |                |          |
 |        |         |    PDU Session ID |                |          |
 |        |         |    DNN: internet  |                |          |
 |        |         |    S-NSSAI: SST=1 |                |          |
 |        |         |    RAT Type: NR   |                |          |
 |        |         |    AccessType:3GPP}|               |          |
 |        |         |                   |--Nudr_DM_Query>|          |
 |        |         |                   |  SM Policy Data|          |
 |        |         |                   |<-Nudr_DM_Resp--|          |
 |        |         |                   |  { Subscribed  |          |
 |        |         |                   |    Session-AMBR|          |
 |        |         |                   |    Default QoS }|         |
 |        |         |                   |                |          |
 |        |         |<--Npcf_SMPolicy---|                |          |
 |        |         |  Control_Create   |                |          |
 |        |         |  Response         |                |          |
 |        |         |  { Session AMBR   |                |          |
 |        |         |    UL=50M DL=200M |                |          |
 |        |         |    Default QoS:   |                |          |
 |        |         |     5QI=9, ARP=8  |                |          |
 |        |         |    PCC Rule-01:   |                |          |
 |        |         |     Flow: any     |                |          |
 |        |         |     5QI=9,Gating:OPEN            |          |
 |        |         |    PolicyCtrlTriggers:            |          |
 |        |         |    [RAT_CHANGE,UE_IP_CH] }        |          |
 |        |         |                   |                |          |
 |        |         |--------N4 Session Establishment Req---------->|
 |        |         |  { PDR: detect UE traffic         |          |
 |        |         |    FAR: forward to internet       |          |
 |        |         |    QER: AMBR UL=50M DL=200M       |          |
 |        |         |    URR: report every 100MB }      |          |
 |        |         |<-------N4 Session Establishment Resp----------|
 |        |         |  { UPF F-TEID allocated }         |          |
 |        |         |                   |                |          |
 |<-------PDU Session Established (UE gets IP, QoS=5QI-9)------------|
```

7. Combined Flow — All 3 Policies

```
REGISTRATION PHASE
■■■■■■■■■■■■■■■■■■■■
UE ■■Reg Req■■■ AMF ■■Npcf_AMPolicyControl_Create■■■ PCF
                 ■■■■■ AM Policy (RFSP, Area restrict) ■■■■
                 ■■■■■ UE Policy (URSP rules) via NAS ■■■UE


PDU SESSION PHASE
■■■■■■■■■■■■■■■■■■■
UE ■■PDU Req■■■ AMF ■■■ SMF ■■Npcf_SMPolicyControl_Create■■■ PCF
                             ■■■■■ SM Policy (PCC Rules, QoS) ■■
                        SMF ■■N4■■■ UPF (enforce rules)


APP SESSION PHASE (Optional — AF-triggered)
■■■■■■■■■■■■■■■■■■■
```